This shows that the choice of integers between two given integers is limited. They are finite in number or may be nothing between them. Let us think what will happen, if we consider rational numbers instead of integers? We will see that we can have many rational numbers between any two rational numbers. There are at least two methods to find more rational numbers between any two rational numbers.

Method of Averages:

We know that that the average of any two numbers always lies at the middle of them. For example, the average of 2 and 8 is $\frac{2+8}{2}=5$ and this 5 lies at the middle of 2 and 8 as shown in the following number line.

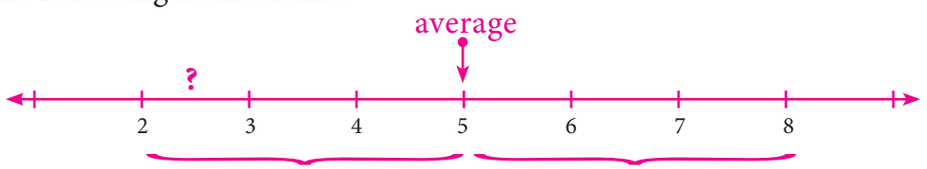


Fig. 1.10

We use this idea to find more rational numbers between any two rational numbers.

Example 1.8

Find a rational number between $\frac{1}{3}$ and $\frac{5}{9}$.

Solution:

The average of $\frac{1}{3}$ and $\frac{5}{9}=\frac{1}{2}\left(\frac{1}{3}+\frac{5}{9}\right)=\frac{1}{2}\left(\frac{1}{3}\times\frac{3}{3}+\frac{5}{9}\right)$ (Why?)

$$=\frac{1}{2}\left(\frac{3}{9}+\frac{5}{9}\right)=\frac{1}{2}\times\frac{8}{9}=\frac{4}{9}$$

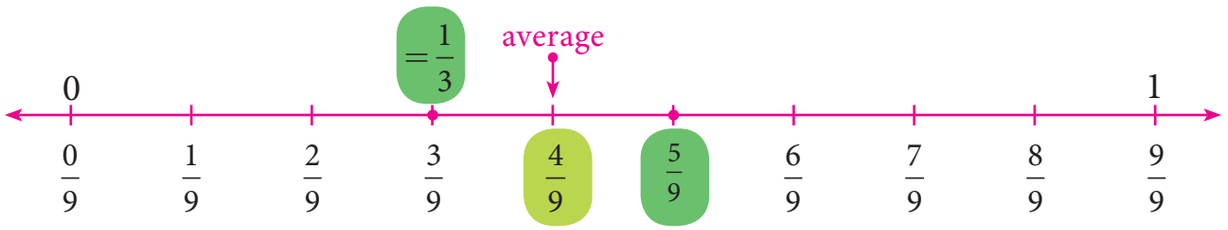


Fig. 1.11

Note that $\frac{4}{9}$ is one rational number we have found in between $\frac{1}{3}$ and $\frac{5}{9}$ and we can find many such numbers in between $\frac{1}{3}$ and $\frac{5}{9}$. This shows that between any two rational numbers there lie an unlimited number of rational numbers! Mathematically, we say that there lie an infinite number of rational numbers between any two rational numbers.

Think



Are there any rational numbers between $\frac{-7}{11}$ and $\frac{6}{-11}$?

Method of Equivalent rational numbers:

We can use the idea of equivalent fractions to get more rational numbers between any two rational numbers. This is clearly explained in the following illustration.

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